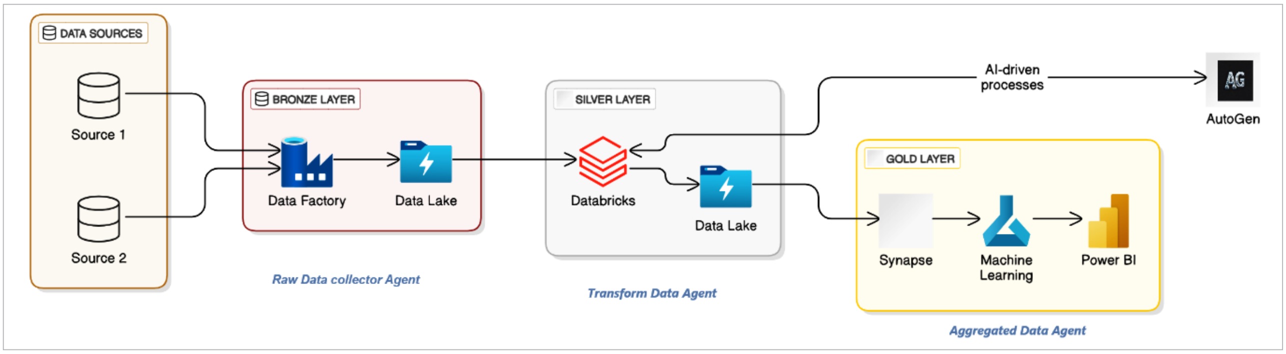

Figure 1: Reference architecture for medallion architecture with Microsoft Autogen

sources without any transformations.

Silver layer: In this layer, data undergoes processing and cleaning. It involves deduplication, normalisation, and validation, ensuring that the data is reliable and usable for further analysis.

Gold layer: This is the final layer, where data is refined and enriched to support advanced analytics and machine learning models. This layer provides high-quality, actionable insights tailored to specific business needs.

This innovative framework can significantly enhance the capabilities of agentic AI solutions, particularly in the financial sector.

Practical use cases in the financial sector

Banking: In the banking sector, medallion architecture can be leveraged to enhance customer service through AI-driven chatbots and virtual assistants. By utilising the bronze layer, raw transactional data can be ingested. The silver layer processes this data to identify patterns, such as frequent transactions or potential fraud. Finally, the gold layer refines this information to provide personalised financial advice and real-time alerts to customers.

Financial services: Financial services can benefit from agentic AI in portfolio management. The bronze layer ingests raw data from stock markets, news feeds, and social media. The silver layer processes this data to filter out noise and normalise disparate datasets. The gold layer uses this refined data to generate insights on market trends and investment opportunities, allowing AI agents to offer customised portfolio recommendations.

Securities: In the securities sector, medallion architecture aids in regulatory compliance and risk management. Raw data from trading systems and regulatory filings is ingested into the bronze layer. The silver layer cleans and validates this data to ensure it meets compliance standards. In the gold layer, AI agents analyse this data to detect anomalies and ensure adherence to regulations, thus mitigating risk.

Capital markets: For capital markets, real-time data processing is crucial. The bronze layer captures real-time trading data. The silver layer processes this data to identify patterns and anomalies, such as unusual trading volumes. In the gold layer, AI agents use this information to predict market movements and inform trading strategies, enhancing decision-making accuracy.

Insurance: In the insurance industry, agentic AI can revolutionise claims processing and fraud detection. The bronze layer ingests raw claims data and policyholder information. The silver layer processes this data to identify inconsistencies and validate claims. In the gold layer, AI agents use the refined data to automate claims processing and flag potential fraudulent activities, improving operational efficiency and reducing costs.

The medallion architecture in Microsoft Autogen offers a robust framework for data-driven decision-making. Its layered approach not only enhances the performance of AI agents but also drives innovation and efficiency across the financial sector.

By embracing medallion architecture, organisations can unlock the full potential of agentic AI, paving the way for smarter, more responsive, and resilient business operations. The future of finance is data-driven, and medallion architecture is a crucial step towards realising this vision. END

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